M4: WORKSPACE & MOTION PLANNING

Week 1 Submission and Weeks 2–5 Roadmap

Project: Two 6-DOF Robotic Arms Playing Chess

Role: M4 — Workspace, Joint Limits, Torque Requirements, Reachability and Trajectory Generation

Purpose

This document defines the Week 1 deliverable, a practical plan for the following four weeks, and the information M4 needs from each teammate. It is designed as a working report: replace every “Pending” item with verified project data as it becomes available.

Important engineering limitation: the task brief does not provide the actual arm dimensions, joint ranges, link masses, centres of mass, gripper mass or motor specifications. Therefore, this document provides valid methods, worksheets and assumptions—not fabricated numerical results.

1. M4 Role and Five-Week Overview

M4 checks whether the arm can reach the chessboard targets, respects mechanical and motion constraints, and can generate usable pick-and-place trajectories. The working dependency chain is: mechanical geometry + board targets → kinematics/IK → reachability and limits → torque/motion constraints → trajectory generation → simulation and hardware validation.

Week	Main work	Expected output
1	Initial workspace calculation; initial joint limits; initial torque requirements with M1/M9.	Assumptions, initial joint-limit table, torque worksheet, blockers, references.
2	Check reachability of the 64 squares and required pick/place poses; set preliminary velocity/acceleration limits; choose trajectory method.	Reachability table/map; identified problem squares; preliminary motion limits and method.
3	Develop trajectory generation and constraint checks; test a pick-and-place sequence.	Trajectory prototype/interface; test cases and logs; constraint validation.
4	Validate complete trajectories, capture moves and safe poses; integrate both arms in simulation.	Normal-move/capture tests; safe poses; integration issue list.
5	Support real pick/place and real chess-square positioning; validate against hardware.	Physical test log; updated target positions and limits; final M4 results.

2. Week 1 — Required Submission

Required format: Research (workspace analysis, joint limits, torque requirements); Workspace (initial calculation, assumptions); Joint Limits (initial limits, reasoning); Torque (initial requirements, inputs from M1, information given to M9); Blockers; References.

2.1 Research — Workspace Analysis

• **Workspace:** the set of positions the robot tool can reach. For chess, also check whether the gripper can reach each target with the required orientation and clearance.

• **Joint limits:** the allowed angular interval for each of the six joints. These must come from the actual mechanical design, not from an assumption that every joint rotates 360 degrees.

• **Torque requirements:** an initial estimate of the torque needed to hold and move the arm, gripper and chess piece. M9 uses it as an input to motor and gearbox selection.

2.2 Workspace — Initial Calculation

First define the board coordinate frame and the robot-base coordinate frame. Each square centre must be expressed relative to the robot base using an agreed board-to-base transform. For a square target, define the required poses: approach above the piece, pick, lift, approach above the destination, place and retreat.

A simplified planar two-link example has approximate radial limits $r_{max} = L1 + L2$ and $r_{min} = |L1 - L2|$. This is only an introductory geometric check; it is not a complete workspace calculation for the actual 6-DOF arm and does not guarantee the required gripper orientation is possible. For the real robot, use M3's FK/IK model and the confirmed joint limits.

Initial assumptions to state explicitly

• The chessboard is stationary during a move.

• The board size, square pitch, board orientation and board-to-base transform are known or explicitly marked provisional.

• M1's latest geometry is authoritative; do not silently replace missing values with guesses.

• A position is considered initially reachable only if IK finds a solution that respects all joint limits and the required tool orientation. Collision-free motion must be checked separately.

2.3 Joint Limits — Initial Table

Ask M1 for the mechanical minimum and maximum angle of every joint, the units, sign convention, and the source (CAD, datasheet or design decision). If the values are not available, mark them "Pending M1"; do not invent values. Later, compare every IK solution with these limits.

Joint	Minimum angle	Maximum angle	Source / notes
J1	Pending M1	Pending M1	Mechanical range; confirm units and sign convention
J2	Pending M1	Pending M1	Mechanical range; confirm units and sign convention

Joint	Minimum angle	Maximum angle	Source / notes
J3	Pending M1	Pending M1	Mechanical range; confirm units and sign convention
J4	Pending M1	Pending M1	Mechanical range; confirm units and sign convention
J5	Pending M1	Pending M1	Mechanical range; confirm units and sign convention
J6	Pending M1	Pending M1	Mechanical range; confirm units and sign convention

Reasoning: an IK solution that asks a joint to move outside its allowed interval is invalid, even if the mathematical solver returns it. If a target fails, check for an alternative IK solution before concluding that the target is unreachable.

2.4 Torque — Initial Requirements

For a preliminary static gravity estimate, use $\tau = m \times g \times r_\perp$, where m is the mass in kilograms, $g \approx 9.81$ m/s², and $r_\perp$ is the perpendicular distance in metres from the joint axis to the load's line of action. Sum the relevant downstream link, gripper and payload contributions for each joint and critical pose. The required torque varies with arm configuration.

• Critical initial pose to inspect: arm extended horizontally, because gravity may create a large moment at shoulder/elbow joints.

• Also inspect the real pick pose, lift pose and any other pose where the load has a large perpendicular moment arm.

• This is a gravity-only estimate, not final motor sizing. Acceleration, friction, transmission efficiency, duty cycle, thermal limits and safety margin must also be considered.

Inputs needed from M1

• Link lengths, joint axes and locations.

• Mass of each link and its centre of mass, preferably from CAD.

• Mechanical joint limits and any gearbox/mechanical restrictions.

Additional inputs needed from M2

• Gripper mass and tool-centre-point location.

• Chess-piece dimensions/mass range and required pick/place heights.

Information to give M9

• A per-joint preliminary gravity torque estimate for named critical poses.

• The masses, centres of mass, lever arms, units, assumptions and source/version used.

• Known speed/acceleration expectations and a clear list of missing data.

M9 should compare the estimate with motor/gearbox torque-speed data, ratio and efficiency, driver/current limits, thermal/duty constraints and safety margin. The motor selection may feed back into the feasible motion limits.

2.5 Week 1 Blockers

• M1 has not supplied confirmed link geometry, joint ranges, masses or centres of mass.

• The board-to-base transform, square pitch, piece heights or gripper dimensions are not confirmed.

• M3's FK/IK model or input/output conventions are not yet agreed.

• M9's motor/gearbox specifications are pending, so torque feasibility and final speed limits cannot yet be confirmed.

2.6 References

• Project source: "Chess Robot Team Structure" / team role and milestone document supplied by the project team.

• Modern Robotics — Task Space and Workspace: https://modernrobotics.northwestern.edu/nu-gm-book-resource/2-5-task-space-and-workspace/

• Modern Robotics — Inverse Kinematics of Open Chains: https://modernrobotics.northwestern.edu/nu-gm-book-resource/inverse-kinematics-of-open-chains/

• Modern Robotics — Chapter 5, velocity kinematics and statics: https://modernrobotics.northwestern.edu/chapters/chapter5/

3. Week 2 — Detailed Plan

Goal: move from initial estimates to systematic reachability and motion constraints.

Step 1: get the latest dimensions, joint axes and limits from M1. Confirm that M3's model uses the same joint order, units, sign convention and base frame.

Step 2: build a target list for all 64 square centres. For each square, include approach, pick, lift, destination approach, place and retreat poses as applicable. Convert targets to the agreed robot-base frame.

Step 3: use M3's IK interface to test targets. Record whether IK succeeds, the returned joint angles, any joint-limit violation, orientation failure and possible collision concerns. A target that is reachable at one orientation may fail at another.

Step 4: review failed or awkward targets with M3 and M1. Check coordinate transforms, alternative IK solutions, singularities, base placement and gripper clearance before calling a square unreachable.

Step 5: agree preliminary per-joint velocity and acceleration limits using motor, gearbox, mechanical and controller constraints. Keep units consistent: rad/s or deg/s, and rad/s² or deg/s².

Step 6: choose a trajectory approach with M5/M6/M10. A trapezoidal velocity profile is a practical point-to-point baseline; cubic or quintic polynomials can be considered when smoother endpoint velocity/acceleration is needed.

Week 2 deliverables: 64-square reachability table/map; failed-target list; preliminary velocity/acceleration limits; trajectory-method choice and rationale.

4. Week 3 — Trajectory Generation

Goal: create a usable, constraint-checked pick-and-place trajectory.

Step 1: agree the trajectory interface with M5, M6 and M10: joint names/order, units, start/target configurations, duration, sample interval, and time-stamped output format.

Step 2: implement or prototype a function that generates joint-space positions over time between a start configuration and target configuration. Ensure it uses the agreed IK solutions.

Step 3: check every trajectory sample against joint-position limits. Estimate or calculate velocity and acceleration and verify they stay within agreed limits. Reject or regenerate invalid trajectories.

Step 4: create the sequence: approach → descend/pick → close gripper → lift → move above destination → descend/place → open gripper → retreat to safe pose. Coordinate gripper timing with the relevant team member.

Step 5: test a small number of square-to-square moves first, then expand. Log duration, peak velocity/acceleration, failed stages and changes made.

Week 3 deliverables: trajectory prototype/interface; constraint checks; pick/place test evidence; handoff notes for M5/M6/M10.

5. Week 4 — Full Sequence and Two-Arm Integration

Goal: validate complete chess moves and prepare the system-level simulation.

Step 1: test a complete normal move as one sequence, including approach, pick, lift, transfer, place, release and retreat. Test several board regions and piece heights.

Step 2: coordinate with M8 on capture moves. If the destination square is occupied, agree the sequence and captured-piece storage location; normally the captured piece must be moved clear before the moving piece is placed.

Step 3: define and test home, waiting, safe-transit, approach and retreat poses. Verify their reachability and check collision clearance with the board, pieces and second arm.

Step 4: integrate with M6/M10 in simulation. Two individually valid trajectories can still collide if both arms execute at the same time, so validate their combined timing and swept paths.

Step 5: log each failure with start/target pose, failed stage, likely cause, owner and next action. Separate planning failures from controller tracking or simulation-model failures.

Week 4 deliverables: normal-move and capture tests; safe-pose definitions; two-arm simulation results; updated constraints and integration issue list.

6. Week 5 — Physical Pick-and-Place

Goal: validate the motion against the assembled hardware and real chessboard.

Step 1: verify actual dimensions, base position, joint zero positions, gripper tool-centre point and mechanical travel with M1/M2/M3/M5/M6. The real assembly may differ from simulation.

Step 2: calibrate board-to-base alignment with M3 and verify a few known square targets before attempting full-board moves.

Step 3: use the team's safety procedure, test emergency stop, start at conservative speed, and test in stages: single joint → single arm → pick/place → square-to-square move → chess move → coordinated two-arm operation.

Step 4: compare commanded and observed motion. If the gripper misses, investigate calibration, kinematic model, joint-zero offsets, flex or controller tracking; document corrections rather than applying unexplained offsets.

Step 5: coordinate with M5/M9 if torque, speed, acceleration or tracking limits prevent the intended motion. Update limits only with a recorded reason and safety review.

Week 5 deliverables: physical pick/place results; verified square targets; updated motion constraints; final test log and remaining issues.

7. What M4 Needs from Each Member

Request each input with units, source/version, and status (measured, CAD-derived, estimated or pending). The exact division of work should follow the team's role sheet; the table below lists dependencies relevant to M4.

Member	Inputs M4 requests	How M4 uses them
M1 — Mechanical design	DOF; link lengths and joint axes; base geometry; mechanical joint limits; link masses and centres of mass; CAD version; mounting and transmission constraints.	Workspace, joint-limit and torque calculations.
M2 — Board / gripper / fabrication	Board dimensions and square pitch; board pose; piece dimensions and mass range; gripper mass/dimensions; tool-centre point; pick/place heights and clearances.	Target coordinates, payload and grasp constraints.
M3 — Kinematics	Coordinate frames; FK/IK model/interface; joint order and sign convention; orientation conventions; alternative IK solutions and singularity notes.	Test target reachability and convert poses to joint configurations.
M5 — Control	Trajectory input format; units and timing; controller limits; measured tracking error; expected speeds/accelerations; joint command conventions.	Make planned trajectories executable and identify control constraints.
M6 — Simulation / ROS integration	URDF/Xacro joint names and limits; inertial/collision model; Gazebo/RViz setup; ROS interfaces; trajectory execution and simulation logs.	Validate reachability, collisions and trajectories in simulation.
M7 — Computer vision	Board detection/orientation; square indexing; calibration and board pose; transform handoff; detection uncertainty.	Provide square identity/pose consistently to the motion pipeline.
M8 — Chess logic	Source and destination square; capture sequence; special-move requirements if in scope; captured-piece storage location agreed by the team.	Translate a chess move into the physical pick/place sequence.
M9 — Electronics and motors	Motor models; torque-speed data; gearbox ratio/efficiency; driver current/voltage limits; encoder/step data; duty/thermal limits and safety margin.	Compare torque estimates and establish feasible joint motion limits.
M10 — System integration	ROS package and interface conventions; frame/joint naming; Robot 1/Robot 2 namespaces; trajectory handoff; status/error handling and sequencing.	Integrate M4 trajectory output with the complete system.

Coordination rule

Agree one definition for coordinate frames, joint names/order, angle units, tool-centre point and trajectory message/file format. If a dependency is missing, log it as a blocker and mark assumptions clearly. Do not present provisional estimates as verified design results.

8. Working Templates

Reachability log

Target	Coordinates/frame	Orientation	IK result	Limits pass?	Collision / notes
e4 approach	Pending board transform	Pending M3	Pending	Pending	Pending test
e4 pick	Pending board transform	Pending M3	Pending	Pending	Pending test

Torque calculation log

Joint / pose	Loads included	Mass, COM, moment arm	Torque	Status
Example: J2, arm horizontal	Downstream links + gripper + piece	Pending M1/M2	To calculate	Preliminary
Other critical pose	Pose-specific loads	Pending	To calculate	Preliminary

Blocker log

Blocker	Owner	Impact	Needed by	Next action
Joint limits missing	M1	Cannot validate IK	Week 2	Request joint table
FK/IK interface missing	M3	Cannot automate target tests	Week 2	Agree input/output
Motor data missing	M9	Torque feasibility provisional	Weeks 2–3	Request datasheets

9. Final Checklist

- ■ Week 1: document workspace method and assumptions.
- ■ Week 1: request six real joint ranges from M1; label missing values Pending.

- ■ Week 1: prepare torque worksheet and share inputs/estimates with M9.
- ■ Week 1: record blockers and references.
- ■ Week 2: test all 64 square targets and required pick/place poses.
- ■ Week 2: record preliminary velocity/acceleration limits and trajectory method.
- ■ Week 3: generate a constrained pick/place trajectory and test it.
- ■ Week 4: validate normal/capture sequences and safe poses in simulation.
- ■ Week 5: support real pick/place and verify board target positions.

Final note

Every result should be traceable to its geometry version, input source, units, assumptions and test evidence. The most important Week 1 achievement is not pretending to have final numbers; it is obtaining the right inputs, applying the correct initial methods, and making missing information visible early.